A
Alignment with null-defining groups
R2
0.0
0.2
0.4
0.6
0.8
Three-spined
(region)
Nine-spined
(lineage)
Populations weighted equally
Individuals weighted equally
B
Alignment with five relationship-matrix axes
R2
0.00
0.25
0.50
0.75
1.00
3sp
raw ecotype
regional null
9sp
raw ecotype
lineage null
9sp
adjusted ecotype
lineage null
9sp
adjusted ecotype
GRM null
GRM-matched continuous null
Within-lineage label permutation
Within-region label permutation